

Blank screen

Perception Model

2nd
2nd

1st

Cor:

$$y = \boxed{50000} * x + \boxed{100000}$$

$$y = 300x_1 + 10000x_2 + 100000 = 110000$$

Ph: 70
Chem: 100
math: 50

Phy 80
chem 85
maths 90 = 255

Cutoff =

240

$$Z = 80x_1 + 85x_2 + 90x_3 - 240$$

$Z \geq 0$
 $Z < 0$

Adms
(No adms 'nbil')

first example
Perceptu

$$(BIS Adm) = 1 \times \text{Phy} + 1 \times \text{chem} + 1 \times \text{maths} - 240$$
$$= 1 \times 70 + 1 \times 100 + 1 \times 50 - 240 = -20 < 0$$

$$Z(DU) = 1 \times \text{phy} + 1 \times \text{chem} + 1 \times \text{mats} + 0.5 \times \text{w} + 0.25 \times \text{phy} \times \text{chem} - 320$$

$$\begin{bmatrix} 80 \\ 50 \\ 10 \\ 150 \\ 10 \end{bmatrix}$$

$$+ (b) \rightarrow +$$

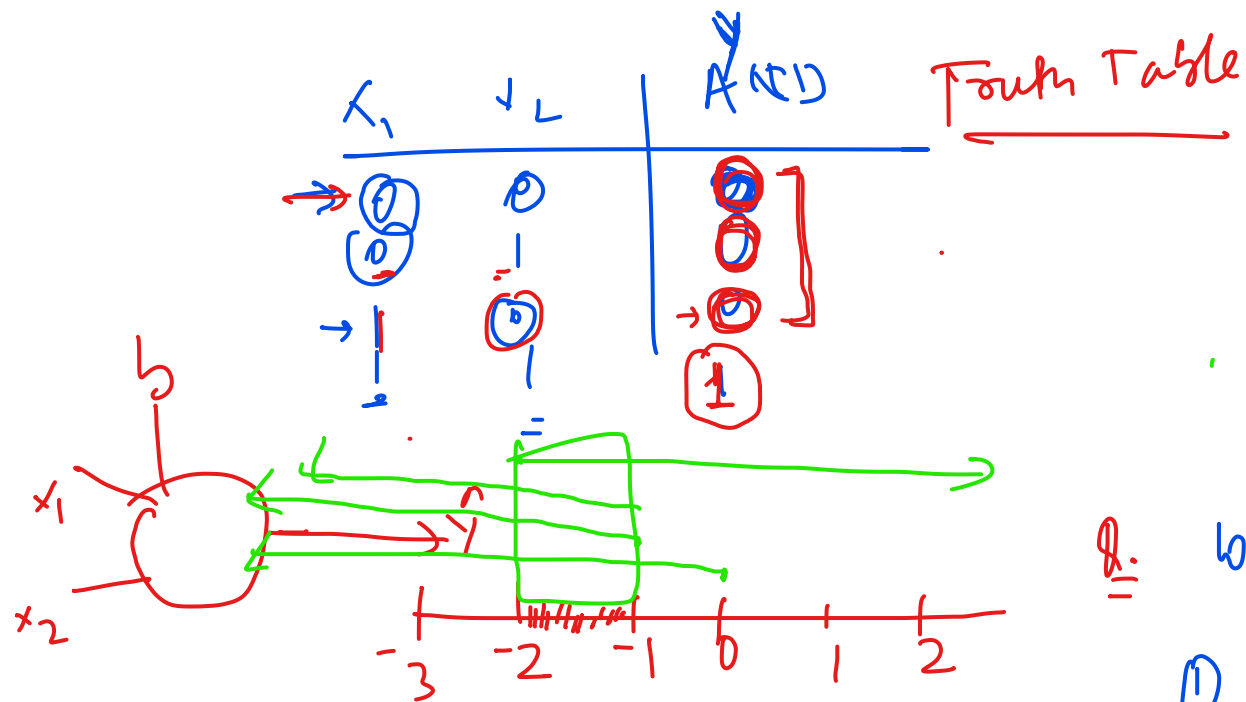
$$= \sum_{i=1}^3 w_i x_i + b$$

$$x_1 = \text{phy}$$

$$x_2 = \text{chem}$$

x_1	x_2
1	1
1	0
0	1
0	0

OR	AND	XOR	Not $[x_1]$	Not $[x_2]$
1	1	0	0	0
1	0	1	0	1
1	0	1	1	1
0	0	0	1	0



$$Z = w_1 x_1 + w_2 x_2 + b$$

$$1 + b = 0$$

$$b = -1$$

$$\hat{y} = \begin{cases} 1 & Z > 0 \\ 0 & Z < 0 \end{cases}$$

Q. $w_1 = 1$ $w_2 = 1$

$$① Z = 1x_1 + 1x_2 + b$$

$$Z = 1 \times 0 + 1 \times 0 + b$$

$$Z = 0 + b < 0$$

$$③ Z = 1x_1 + 1x_0 + b$$

$$Z = 1 + b < 0$$

$$b < -1$$

what values of b should keep

$$② Z = 1x_0 + 1x_1 + b$$

$$Z = 1 + b < 0$$

$$b < -1$$

$$④ Z = 1x_1 + 1x_1 + b$$

$$Z = 2 + b > 0$$

$$b > -2$$

$$-2 < b < -1$$

$$b = -1.5$$

$$b = -1.6$$

$$b = -1.7$$

$$b = -1.8$$

$$b = -1.9$$

$$2 + x < 0$$

$$x < -2$$

$$1 + b < 0$$

$$b < -1$$

Q.

f_1 x_1	f_2 x_2	<u>Class</u> y
0	0	0
0	1	1
1	0	1
1	1	1

$$b = -1.5$$

$$w_1 = 1$$

$$w_2 = 1$$

$$\begin{bmatrix} x_1 & x_2 \\ 0 & 0 \\ 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}$$

$$Z = w_1 x_1 + w_2 x_2 + b$$

$$Z = 1 \times x_1 + 1 \times x_2 + (-1.5)$$

$$Z = 0 + 0 - 1.5$$

$$Z = -1.5 < 0 \quad \hat{y} = 0$$

$$Z = 1 \times 1 + 1 \times 1 - 1.5$$

$$= 2 - 1.5$$

$$= 0.5 > 0 \quad \hat{y} = 1$$

y = Actual
 $\hat{y}$ = Predicted value

$$Z = 1 \times 0 + 1 \times 1 - 1.5$$

$$= 1 - 1.5$$

$$= -0.5 < 0$$

$$\hat{y} = 0$$

$$w_1 = 1 \quad w_2 = 1 \quad b = ?$$

$$\hat{y} = \begin{cases} 1 & Z > 0 \\ 0 & Z < 0 \end{cases}$$

$$1) Z = w_1 x_1 + w_2 x_2 + b$$

$$= 1 \times 0 + 1 \times 0 + b$$

$$= \boxed{b < 0}$$

$$(2) \quad Z = |x_0| + |x_1| + b$$

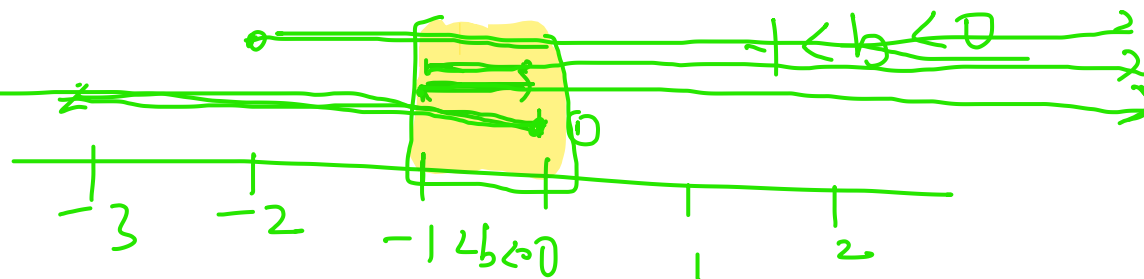
$(1+b > 0)$ $b > -1$

$$(3) \quad Z = |x_1| + 0|x_1| + b > 0$$

$(1+b > 0)$
 $b > -1$

$$(4) \quad Z = |x_1| + |x_2| + b$$

$(2+b > 0)$
 $b > -2$



$$b = -0.5$$

$$b = -0.6$$

$$b = -0.7$$

$$\begin{array}{cc|c} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{array}$$

$$Z = |x_1| + |x_2| - 0.7$$

$$\begin{array}{cc|c} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{array}$$

$$Z_0 = w_0 b + w_1 x_1 + w_2 x_2 \quad \begin{array}{l} w_1 = 0.75 \\ w_2 = 0.75 \end{array}$$

$$Z = 1b + w_1 x_1 + w_2 x_2$$

$$\hat{y} = \begin{cases} 1 & Z > 0 \\ 0 & Z \leq 0 \end{cases}$$

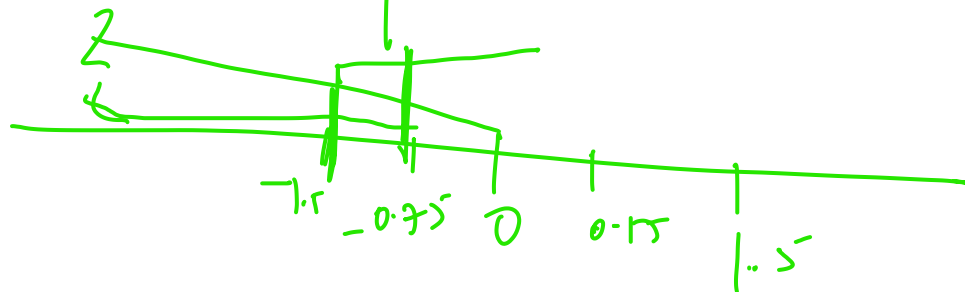
$$Z = b + 0.75x_1 + 0.75x_2$$

pt

$$\begin{array}{l} Z = b + 0 < 0 \\ \underline{b < 0} \end{array}$$

$$\begin{array}{l} Z = b + 0.75x_1 + 0.75x_2 \\ Z = b + 0.75 < 0 \\ \underline{b < -0.75} \end{array}$$

$$\begin{array}{l} Z = b + 0.75x_1 + 0.75x_2 \\ Z = b + 1.5 < 0 \\ \underline{b < -1.5} \end{array}$$



$$\boxed{-1.5 < b < -0.75}$$

$$\boxed{b = -0.75}$$

$$w_1 = 0.75$$

$$w_2 = 0.75$$

$$b = -0.85$$

$$w_1 = 0.75$$

$$w_2 = 0.75$$

$$b = -0.80$$

f_1	f_2	y
0	0	0
0	1	0
1	0	0
1	1	1

$$w_1 = 1$$

$$w_2 = 1$$

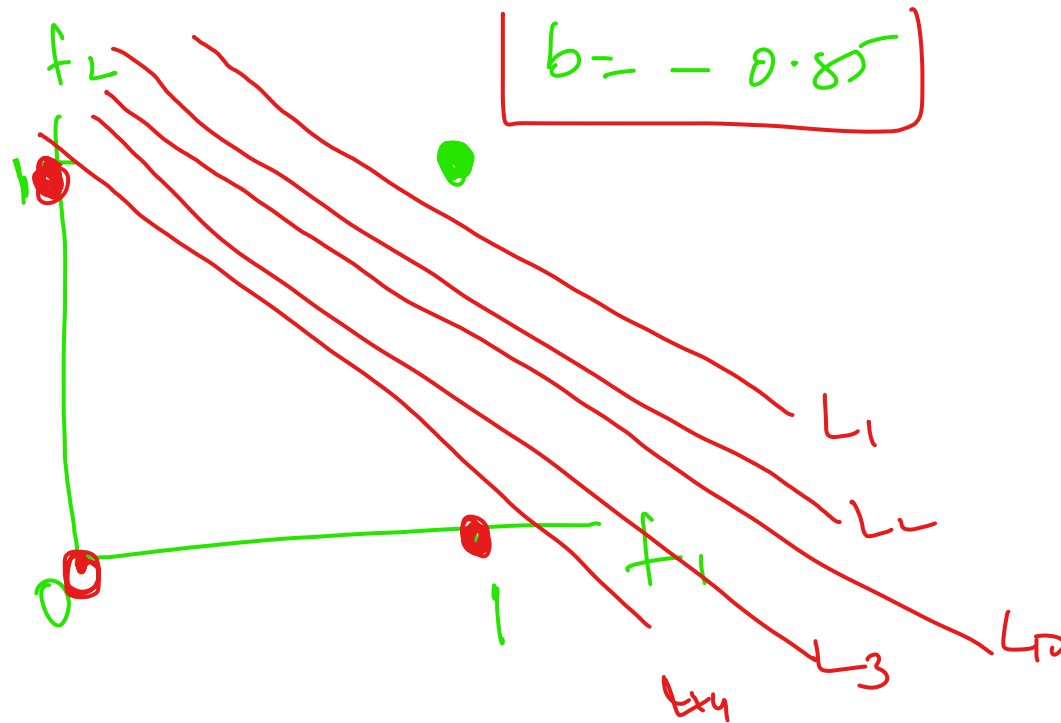
$$b = -1.5$$

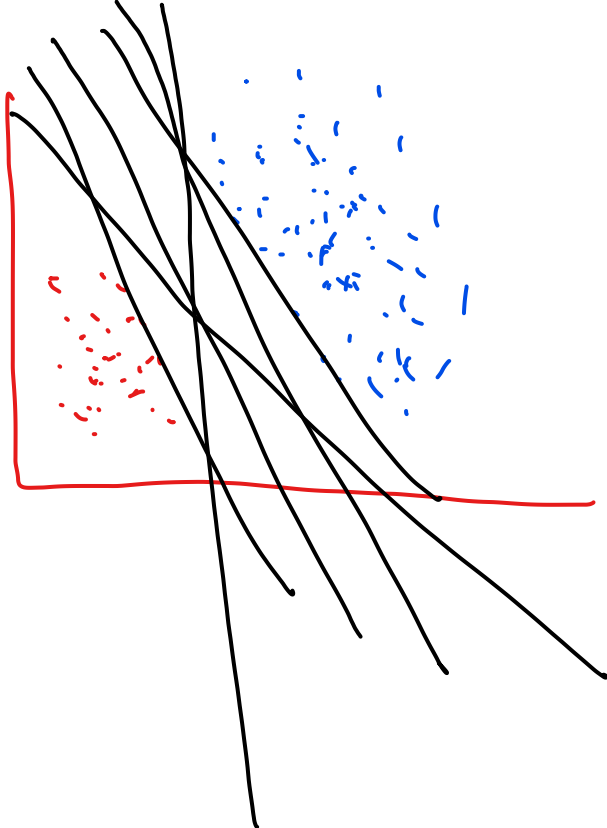
$$w_1 =$$

$$w_2 =$$

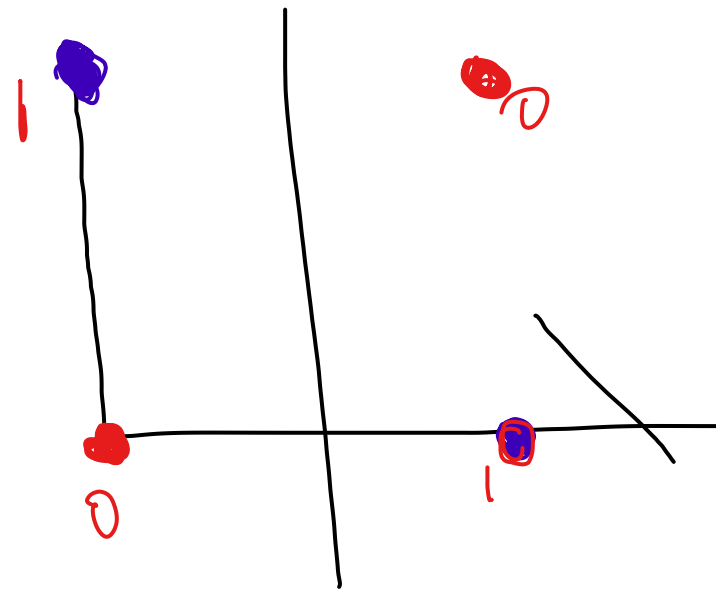
$$b =$$

Gate





w_1
 w_2
 b



Perceptron can not
 solve XOR

